Implementation of Fully Connected Feed Forward Neural Network (FCFNN) using TensorFlow.Keras

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1 Introduction

In this assignment, a Fully Connected Feed Forward Neural Network (FCFNN) was designed and implemented using TensorFlow.Keras. The architecture was selected according to personal preference and manually drawn before implementation.

2 Network Diagram

Figure 1 shows the manually drawn Fully Connected Feed Forward Neural Network architecture consisting of 8 input neurons, three hidden layers (4, 8, 4 neurons respectively) and 10 output neurons.

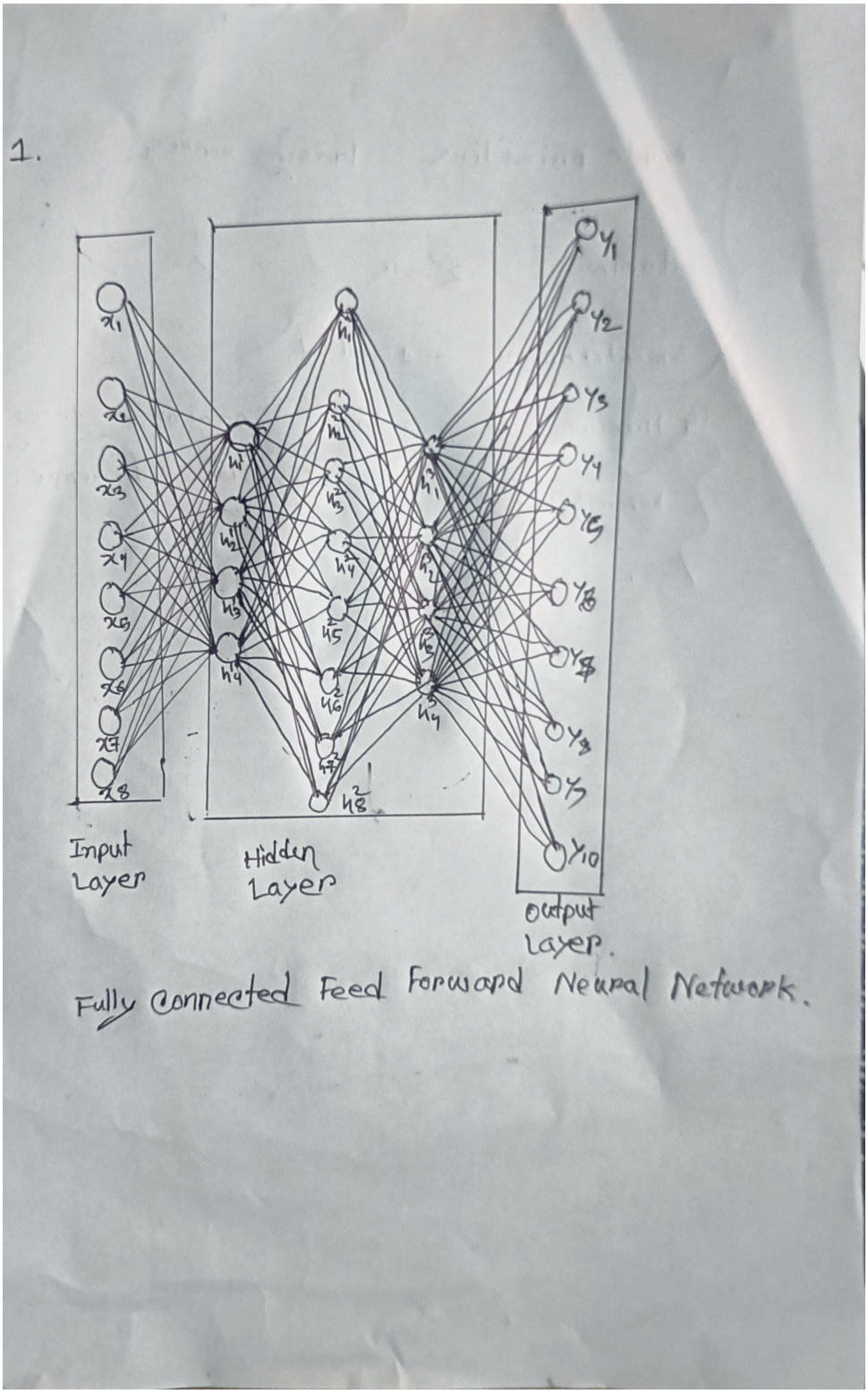

Figure 1: Manually Drawn FCFNN Architecture (8-4-8-4-10)

3 Network Architecture

The proposed architecture consists of:

- Input Layer: 8 neurons

- Hidden Layer 1: 4 neurons
- Hidden Layer 2: 8 neurons
- Hidden Layer 3: 4 neurons
- Output Layer: 10 neurons

ReLU activation function was used in all hidden layers and Softmax activation function was used in the output layer for multi-class classification.

4 Parameter Calculation

$$\begin{aligned} \text{Layer 1: } & 8 \times 4 + 4 = 36 \\ \text{Layer 2: } & 4 \times 8 + 8 = 40 \\ \text{Layer 3: } & 8 \times 4 + 4 = 36 \\ \text{Output: } & 4 \times 10 + 10 = 50 \end{aligned}$$

$$\textbf{Total Trainable Parameters} = 162$$

The calculated total number of trainable parameters is 162, which matches the model summary produced by TensorFlow.

5 Implementation using TensorFlow.Keras

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import numpy as np

# Dummy dataset (1000 samples, 8 features)
X = np.random.rand(1000, 8)

# 10 class categorical labels
y = tf.keras.utils.to_categorical(
    np.random.randint(10, size=1000),
    num_classes=10
)

# Train-test split
X_train, X_test = X[:800], X[800:]
y_train, y_test = y[:800], y[800:]

# Model Architecture (8-4-8-4-10)
model = Sequential([
    Dense(4, activation='relu', input_shape=(8,)),
    Dense(8, activation='relu'),
    Dense(4, activation='relu'),
```

```
    Dense(10, activation='softmax')
])

# Compile model
model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Show model summary
model.summary()

# Train model
history = model.fit(
    X_train, y_train,
    epochs=50,
    validation_split=0.2
)

# Evaluate model
test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test Accuracy:", test_acc)
```

6 Results

The model summary confirms that the total number of trainable parameters is 162. The model was trained for 50 epochs using the Adam optimizer and categorical cross-entropy loss function. The training and validation accuracy were observed during training.

7 Conclusion

The Fully Connected Feed Forward Neural Network (FCFNN) was successfully designed, manually drawn, and implemented using TensorFlow.Keras. The parameter calculation was verified using the model summary. This assignment demonstrates the practical implementation of a multi-layer feed-forward neural network.